U-W-00140a(DSA-DM)
14 June 1967
Interim Revision of
Fed. Spec. U-W-140
16 March 1953

INTERIM FEDERAL SPECIFICATION

WAX, DENTAL, CASTING

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification U-W-140.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers sheet casting wax and ready-made wax shapes suitable for use in dental casting procedures.

1.2 Classification.

1.2.1 Type, grade, and classes. Casting wax covered by this specification shall be of one type, grade, and of the following classes, as specified (see 6.1):

Class A. 28-gage sheet
Class B. 30-gage sheet.
Class C. Ready-made shapes and gages as specified in 3.2.3.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards of the issue in effect on date of invitation for bids, or request for proposal, form a part of the specification to the extent specified herein:

FSC 6520

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SPECIFICATIONS

Federal

MMM-A-250	Adhesive, Water-Resistant, (For Sealing Fiberboard Boxes).
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

STANDARDS

Federal

FED-STD-123	Marking for Domestic Shipment (Civilian Agencies).

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D.C.

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(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

(Copies of Military specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E28-58T Method of Test for Softening Point by Ring-And-Ball-Apparatus.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 Material. Casting wax shall be a uniform mixture of waxes, resins, and hydrocarbons of the paraffin series, combined in such proportions as will result in a material possessing the characteristics specified herein.

3.2 Description. The casting wax shall be suitable for use in dental casting procedures.

3.2.1 Class A. Class A casting wax shall be pink, and shall be furnished in sheets 3-1/2 ± 1/16 inches square and 0.0126 ± 0.0010 inch thick.

3.2.2 Class B. Class B casting wax shall be green, and shall be furnished in sheets 3-1/2 ± 1/16 inches square and 0.0100 ± 0.0010 inch thick.

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3.2.3 Class C. Class C casting wax shall be blue, and shall be furnished in lengths of 4 ± 1/16 inches. Shapes and gages of the wax shall be as specified in Table I. Unless otherwise specified, quantities of the wax shall be furnished in the assortment specified in Table I. The quantity for any one gage and shape shall be within ± 10 percent of the quantity specified in the table. A minimum of 330 of the ready-made shapes constitutes one assortment.

Table I. Class C Wax.

Shape	B & S Commercial Gage *	Diameter, Inch **	Assortment Quantity
Round	18	0.0403	50 each
Round	14	0.0641	50 each
Round	10	0.1019	30 each
1/2 Round	14	0.0641	50 each
1/2 Round	12	0.0808	65 each
1/2 Round	8	0.1285	40 each
1/2 Round	6	0.1620	25 each
1/2 Pear	6	0.1620	20 each

* For information only.
** A tolerance of ± 10 percent in the diameter will be permitted.

3.3 Physical properties. Casting wax shall possess the physical properties specified in 3.3.1 through 3.3.5, inclusive.

3.3.1 Working properties. The wax shall be pliable and readily adaptable at temperatures of 40.0 to 45.0 degrees Centigrade (° C.), and shall copy accurately the surface against which it is pressed. The wax shall cool and harden quickly, shall not be brittle, and shall be easy to trim with a sharp instrument.

3.3.2 Softening point. The softening point of the wax shall be from 60.0 to 70.0° C., when tested as specified in 4.4.1.

3.3.3 Flow. The maximum flow of the wax at 35.0° C. shall be 10.0 percent, and the minimum flow at 38.0° C. shall be 60.0 percent, when tested as specified in 4.4.2.

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3.3.4 Breaking point. The wax shall not fracture, when tested as specified in 4.4.3.

3.3.5 Residue. The melted wax shall leave no solid residue in excess of 0.10 percent of the original weight of the specimen, when tested as specified in 4.4.4.

3.4 Lot or batch number. Each unit package shall be marked with the lot or batch number and date of manufacture.

3.5 Stability. The wax shall conform to the requirements of this specification for not less than two years from date of delivery to the government, when stored in a cool, dry place (50 to 80 degrees Fahrenheit, ^{o} F)

3.6 Delivery schedule. Not more than two months shall have elapsed from date of manufacture of the wax to date of delivery to the Government.

3.7 Workmanship. Workmanship shall be first-class throughout. The wax shall be free from defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

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4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

4.2.2 For tests. Sampling for tests shall be as indicated in Table II. The unit of product for test purposes shall be one package of casting wax.

Table II. Test Criteria

Characteristic	Reqmt.	Test procedure	Unit of product to be tested per lot or batch number	Number of determinations per unit of product
Working properties	3.3.1	4.4.5	2	1*
Softening point	3.3.2	4.4.1	1	2**
Flow	3.3.3	4.4.2	2	3**
Breaking point	3.3.4	4.4.3	2	2*
Residue	3.3.5	4.4.4	1	2**

* Two determinations shall be made for each shape and gage furnished. If any determination fails the requirement, the material shall be rejected.

** If the average of the determinations per unit of product for an applicable test falls below the requirement, the material shall be rejected.

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4.2.3 Sample unit usage. Where feasible, the same sample unit may be used for the determination of two or more test characteristics.

4.3 Examination. The wax shall be examined to determine compliance with the requirements contained in this specification.

4.4 Tests.

4.4.1 Softening point (ring and ball method). The softening point of the wax shall be determined by ASTM method E28-58T.

4.4.2 Flow.

4.4.2.1 Preparation of specimens. A quantity of wax shall be broken into pieces and placed in a metal pouring pan. The pan shall be placed on a surface which is 130 millimeters (mm.) below a 250-watt infrared lamp. The wax, while being stirred, shall be allowed to reach a temperature of 75.0 ± 5.0° C. and be maintained at this temperature until the sample is melted throughout. A thermometer shall be used to measure the temperature. The melted wax shall than be poured into a mold that has been lubricated with a silicone grease whose melting point is higher than that of the melted wax. The mold shall consist of a stainless steel plate 6.0 mm. thick, having flat and parallel top and bottom surfaces, and containing four holes 10.0 mm. in diameter. The axes of the holes shall be perpendicular to the surfaces of the plate. The sides of the holes shall be finished smooth. The mold shall be preheated to a temperature of 55.0 ± 5.0° C., and placed on a smooth glass slab 150 mm. long, 75 mm. wide, and 20 mm. thick, preheated to the same temperature. As the wax freezes and a shrinkage void appears, liquid wax shall be added. When the wax has lost its "mirror-like" surface, a smooth, flat tin-foil or aluminum-foil covered glass plate, preheated to 55.0 ± 5.0° C., shall be placed on the top of the mold. A load of 9,000 grams shall be applied to the top of the foil-covered glass plate for 30 minutes. The weight and the glass plate shall then be removed and the excess wax trimmed away. This may be accomplished by drawing the specimen flush with the surface. The mold shall be removed from the glass slab by gently tapping the side of the mold. The specimens of wax shall be removed from the mold by chilling in water at 10.0° C., and shall be stored at 23.0 ± 2.0° C. for 24 hours before testing.

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4.4.2.2 Test method. The initial length of the specimen shall be determined at 23.0 ± 2.0° C. using a metric micrometer caliper. Four measurements shall be made around the circumference and one measurement shall be made in the center of the specimen. The measurements shall be averaged and recorded to the nearest 0.005 mm. The specimen and flow testing instrument shall be placed in a water bath and held at the testing temperature for 20 minutes prior to testing. The flow testing instrument shall consist of a weight, a shaft having low thermal conductivity, and a brass platen. The total weight in air of these three components shall be 2,000 grams. The weight shall be separated a minimum distance of 75 mm. from the brass platen by the nonconducting shaft. The shaft shall be of hard rubber or a similarly poor thermal conductor to reduce loss of heat from the specimen. The diameter of the brass platen shall be not less than 50 mm. The thickness shall not exceed 6.35 mm. The temperature of the bath shall be controlled to within ± 0.1° C. of the required temperature, a calibrated thermometer being used for determining the temperature. Agitation shall be provided in the water bath by means of a mechanical stirrer. A thin sheet of waterproof cellophane shall be placed between the instrument and each end of the specimen. The bottom of the specimen shall be 50 mm. below the surface of the water in the bath. A constant axial load of 2,000 grams shall then be applied to the specimen for 10 minutes, after which the specimen shall be removed and cooled in air to 23.0 ± 2.0° C. The cellophane shall be stripped off and the final length determined in the same manner as the original length. The flow, as evidenced by the change in length, shall be reported as percentage of the initial length. The value for flow shall be the average value for three specimens and shall be reported to the nearest 0.1 percent.

4.4.3 Breaking point. A sample of the casting wax at 23.0 ± 1.1° C. shall be bent for 10 seconds at a uniform rate until it is bent double, with the two halves of the specimen in contact for their entire length.

4.4.4 Residue. Approximately one gram of wax shall be placed in a crucible previously conditioned to constant weight by repeated heating to 500° C. and cooling to 23.0 ± 2.0° C. The conditioned, tared, and loaded crucible shall be transferred to a furnace at 23.0 ± 2.0° C. The temperature of the

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furnace shall be increased to 500° C. and maintained at this temperature for one hour. The crucible shall then be removed from the furnace, placed in a desiccator, and allowed to cool to 23.0 ± 2.0° C. before weighing. The value for residue shall be the average value of two determinations and shall be reported to the nearest 0.02 percent.

4.4.5 Working properties. The casting wax shall be given a practical performance test to determine compliance with the specified working properties cited in 3.3.1.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One box containing an assortment of 330 shapes, or one box containing 32 sheets as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table III shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table III. Packaging Quantities

Class	Unit package	Intermediate package	Exterior container
A	1 unit	12 units	72 units
B	1 unit	6 units	72 units
C	1 unit	12 units	144 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

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5.1.3 Level A.

5.1.3.1 Unit package.

5.1.3.1.1 Class A and B. Thirty-two sheets of wax shall be interleaved with glazed or other suitably coated paper and packaged in a carton of appropriate size constructed in accordance with PPP-B-676, except that commercial colors are acceptable. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of or damage to the contents. Closure of carton shall be adequate to prevent spilling of contents under normal handling.

5.1.3.1.2 Class C. One assortment of 330 wax shapes shall be packaged in a carton of appropriate size conforming to PPP-B-676, except that commercial colors are acceptable. Carton shall be suitably partitioned so that each shape is contained in an individual compartment. Each compartment shall be identified as to shape, gage, quantity, and diameter. Closure of carton shall be adequate to prevent spilling of contents under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design, constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2-inch wide tape conforming to PPP-T-76. Closing of bottom flap with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type I, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant. Closure and strapping shall be in accordance with appendix of applicable box specification.

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5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic. Closure shall be in accordance with appendix of applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civilian agencies. Shipments shall be marked in accordance with FED-STD-123.

5.3.2 Military agencies.

5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number, name of contractor, lot (control) number, date of manufacture in lieu of date packed and the following legend shall be shown:

"STORE IN A COOL PLACE (50° - 80° F.)"

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number, date of manufacture in lieu of date packed, and the following legend shall be shown:

"STORE IN A COOL PLACE (50° - 80° F.)."

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Class required (see 1.2.1).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements .

SPECIFICATION

U-W-00110a(DSA-DM)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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